



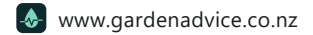
DEVICE SETUP GUIDE

Garden Monitor

Sensors, Watering & Factory Reset Reference

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This guide covers how to add sensors and watering relay systems to your Garden Monitor, and how to factory reset each device if needed.



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System Overview

Your Garden Monitor system consists of:

Device	Purpose
Garden Monitor	The central hub. Connects to your WiFi and provides the web interface for controlling everything.  www.gardenadvice.co.nz
Indoor Sensor	Monitors soil moisture, temperature, and humidity for indoor plants.
Outdoor Sensor	Monitors soil moisture for outdoor garden beds.
Relay Watering System	Controls up to 4 watering zones via solenoid valves or pumps.

The Garden Monitor communicates wirelessly with sensors and watering systems using a long-range radio (LoRa). All sensors and watering systems must be **paired** to your Garden Monitor before they can be used.

Setting Up the Garden Monitor

The first time you power on the Garden Monitor — or after a factory reset — it has no WiFi details stored, so it creates its own temporary setup hotspot and walks you through a 3-step wizard.

Step 1 — Connect to the setup hotspot

1. Power on the Garden Monitor and give it about a minute to start up.
2. On your phone or computer, open your WiFi settings and connect to the network named **GardenMonitor-XXXXXX** (the last part is the device's own network ID).
 - This is an open network — there is no password.
3. Open a web browser and go to **<http://192.168.4.1>**. The setup wizard opens at **Step 1 of 3**.

If the wizard does not open on its own, go to <http://192.168.4.1/setup> directly.

Step 2 — Name the device

Enter a name to identify this Garden Monitor, for example "Back Garden Monitor", then tap **Next**. The name is shown on the device's own display.

Step 3 — Set the date and time

1. Choose your **Date Format**:
 - **DD/MM/YYYY** — NZ, Australia, UK, Europe
 - **MM/DD/YYYY** — United States
2. Choose your **Timezone** from the list (for example, New Zealand UTC+12/+13).
3. Tap **Next**.

Step 4 — Connect to your WiFi

1. Tap **Scan** to list the networks in range, then pick yours from the dropdown — or type in your **WiFi Name (SSID)** yourself.
2. Enter your WiFi **Password**. Leave it blank for an open network.
3. Tap **Save & Connect**.
4. The Garden Monitor saves the details and restarts. Your phone will drop off the setup hotspot, so reconnect it to your normal WiFi network.

If the Garden Monitor cannot reach your WiFi, it brings the setup hotspot back up and quietly retries your stored network every 5 minutes. If your router was simply switched off, it will reconnect on its own.

Step 5 — Set your admin password

1. Once the Garden Monitor has restarted, open **http://gardenmonitor.local** in a browser on the same WiFi network.
2. The dashboard opens without a password. The first time you go to **Settings**, you are asked to create an admin password.
3. Enter a password of at least 4 characters, confirm it, and click **Set Password & Continue**.
4. From then on, the settings pages ask you to sign in. The username is **Admin**, and the password is the one you just set.

Keep this password somewhere safe. If you lose it, the only way back in is a factory reset of the Garden Monitor, which erases all configuration.

Accessing the Garden Monitor

1. Make sure your Garden Monitor is powered on and connected to your WiFi network.
2. On a device connected to the same WiFi network, open a web browser.
3. Go to: **http://gardenmonitor.local**
 - If this doesn't work, use the IP address shown on the Garden Monitor's display instead (e.g. `http://192.168.1.50`).
4. The Garden Monitor dashboard will load. From here, navigate to **Settings** to manage devices.

Adding an Indoor or Outdoor Sensor



Step 1 — Power on the sensor

Insert the batteries (or connect power) to the sensor. On the outdoor sensor, also set the **On/Off** switch inside the enclosure to **On**. The sensor will automatically begin broadcasting to the Garden Monitor, and will appear in the **Pending Devices** tab within a minute or two.

Tip: Keep the sensor within a reasonable distance of the Garden Monitor during setup to ensure a reliable signal.

Step 2 — Pair the sensor

1. In the web interface, go to **Settings**.
2. Click the **Pending Devices** tab.
3. Your sensor will appear in the **Pending Sensors** list, showing its device ID and current readings.
4. Click **Pair & Activate**.
5. The Garden Monitor will send a pairing request to the sensor. Wait up to 30 seconds for the pairing to complete.
6. Once paired, the sensor will automatically enter **calibration mode** — the sensor display will update to show a calibration link.

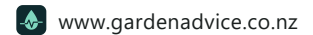
If the sensor does not appear after a few minutes, check the battery and make sure you are within radio range of the Garden Monitor.

Step 3 — Calibrate the sensor

Calibration teaches the sensor what "dry" and "wet" look like for your specific soil. You will need to do this once per sensor.

1. After pairing, click the **Calibrate** link shown next to the sensor.
2. **Dry calibration:**
 - Hold the sensor probe in the **air** (not touching anything).
 - Wait for the live reading to appear and stabilise.
 - Click **Set Dry Value**.
3. **Wet calibration:**
 - Insert the sensor probe into a **glass of water**.
 - Wait for the live reading to appear and stabilise.
 - Click **Set Wet Value**.

4. Calibration is now complete. The sensor will be added to your active sensors list and the page will redirect automatically.



After wet calibration, dry the sensor probe before placing it in soil.

Step 4 — Configure the sensor

1. Go to **Settings** → **Sensors** tab.
2. Find your newly added sensor and set:
 - **Name** — a descriptive name (e.g. "Herb Garden", "Living Room Fiddle Leaf")
 - **Location** — Indoor or Outdoor
 - **Soil Type** — Regular Soil or Potting Mix
 - **Moisture Preference** — Dry, Normal, or Wet (based on your plant's needs)
 - **Garden** — assign it to one of your gardens if you have multiple set up
3. Your sensor will now appear on the dashboard and begin reporting readings.

Adding a Relay Watering System

The Relay Watering System controls up to **4 watering zones** (relays). Each zone can be independently named, enabled, and assigned to a garden.

Step 1 — Power on the watering system

Connect power to the Relay Watering System. It will immediately begin broadcasting to the Garden Monitor and will appear in the **Pending Devices** tab within a minute or two.

Make sure your solenoids or pumps are **not** connected to the relay outputs during initial setup. Connect them after pairing and configuration is complete.

Step 2 — Pair the watering system

1. In the web interface, go to **Settings**.
2. Click the **Pending Devices** tab.
3. Your watering system will appear in the **Pending Watering Systems** list, showing its device ID.
4. Optionally, enter a **name** for the system before pairing.
5. Click **Pair & Adopt**.
6. The Garden Monitor will pair with the watering system. This is usually completed within a few seconds.
7. Once paired, the relay system will appear as active on your dashboard.

Step 3 — Configure the relays



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1. Go to **Settings** → **Watering** tab.
2. Find your newly paired watering system.
3. For each of the 4 relay channels, you can set:
 - **Relay Name** — e.g. "Front Lawn", "Veggie Bed", "Drip Line"
 - **Garden** — assign the relay to one of your gardens
 - **Enable/Disable** — toggle individual relays on or off
4. Your watering system will now be controllable from the dashboard. You can manually trigger each zone and set durations.

Factory Resetting Devices

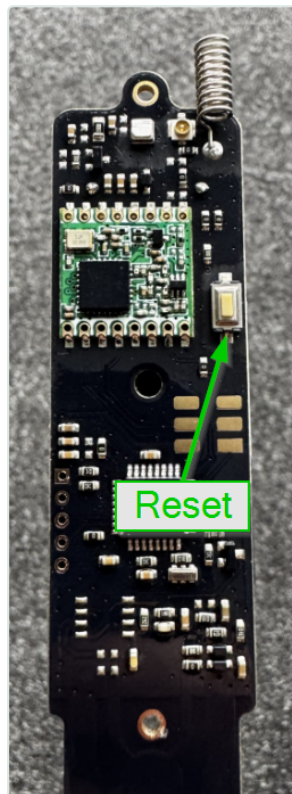


A factory reset clears the pairing information from a device, returning it to a state where it can be re-paired. Use this if you need to move a device to a different Garden Monitor, or if a device is behaving incorrectly.

Note: A factory reset only clears pairing — it does not change any hardware settings.

Factory Reset — Indoor Sensor

The indoor sensor has a single **Reset** button on the board, next to the radio module.



Indoor sensor — the Reset button sits beside the radio module.

1. **Locate the Reset button** on the sensor board (shown above).
2. **Press it 3 times in quick succession** — around a second apart, letting the sensor start up between presses.
3. On the third press the sensor clears its pairing and returns to unpaired discovery mode.
4. Confirm it worked: the sensor reappears in the **Pending Devices** tab on the Garden Monitor within a minute or two.

Timing is important. Each press must come within about 30 seconds of the previous one. Once the sensor completes a full reading cycle it clears the counter, and you will need to start again.

The indoor sensor has no status LED, so there is no confirmation on the device itself. Check the **Pending Devices** tab to confirm the reset worked.

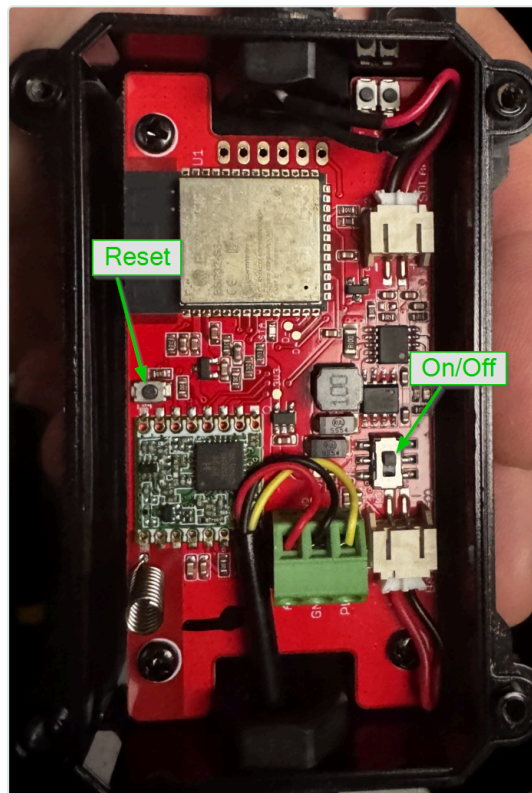
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Taking the batteries out and putting them back counts the same way as a button press, so the older power-cycle method still works if you cannot reach the button.

Alternatively, if the sensor is already paired and within radio range, the Garden Monitor can send a remote reset. Go to **Settings** → **Sensors**, find the sensor, and click **Reset**.

Factory Reset — Outdoor Sensor

The outdoor sensor has two controls inside its enclosure: an **On/Off** switch and a **Reset** button.



Outdoor sensor — Reset sits beside the processor, the On/Off switch on the right.

1. Open the enclosure and locate the **On/Off** switch and the **Reset** button (shown above).
2. Switch the sensor **Off**.
3. **Hold the Reset button down**, and while still holding it, switch the sensor **On**.
4. **Keep holding for 3 seconds**. The status LED stays solid on while the button is held.
 - Release the button any time before 3 seconds to cancel the reset. The LED goes out and the sensor starts up normally.
5. After 3 seconds the LED flashes rapidly 6 times to confirm. The sensor clears its pairing and restarts in unpaired discovery mode.
6. Confirm it worked: the sensor reappears in the **Pending Devices** tab on the Garden Monitor within a minute or two.

The LED also blinks 3 times every time the sensor starts up. That is the normal start-up signal, not the reset confirmation — the reset confirmation is 6 fast flashes after the full 3-second  www.gardenadvice.co.nz

Alternatively, if the sensor is already paired and within radio range, the Garden Monitor can send a remote reset. Go to **Settings** → **Sensors**, find the sensor, and click **Reset**.

Factory Reset — Relay Watering System

The relay watering system has a **factory reset button** (a push button connected between pin 8 and ground on the device board).

1. **Locate the reset button** on the relay board.
2. **Hold the reset button down** while powering the system on.
3. The LED on the device will **blink 8 times** to confirm the reset was successful.
4. The watering system will restart in unpaired discovery mode, ready to be re-paired.

The device's unique ID (e.g. WS04821) is never erased — only the pairing information is cleared.

Factory Reset — Garden Monitor (Hub)

Warning: This will erase **all** configuration including paired sensors, watering systems, gardens, WiFi settings, and historical data. This cannot be undone.

Method A — Web interface (recommended):

1. Go to **Settings** → **System** tab.
2. Scroll to the **Factory Reset** section.
3. Click **Factory Reset Device**.
4. Confirm the action in the dialog box.
5. The Garden Monitor will restart with factory defaults.

Method B — Hardware button:

1. Locate the **BOOT button** on the Garden Monitor board.
2. While the device is powered on, **hold the BOOT button for 5 seconds**.
3. Release the button when prompted (serial console shows countdown).
4. The device will restart with factory defaults.

After a factory reset, you will need to reconnect the Garden Monitor to your WiFi and re-pair all sensors and watering systems from scratch.

Troubleshooting



Problem	What to try
Sensor doesn't appear in Pending Devices	Check batteries. Move sensor closer to the Garden Monitor. Wait 2–3 minutes.
Pairing times out	Ensure the sensor is within range. Try removing and reinserting batteries to restart the sensor, then try again.
Calibration link doesn't appear after pairing	Refresh the Settings page. If the sensor shows as "Calibrating", click directly on its ID to open the calibration page.
Calibration page shows no live readings	The sensor may have lost radio contact. Click Retry on the calibration page.
Watering relay doesn't appear in Pending Devices	Check power to the relay board. Move it closer to the Garden Monitor.
Relay not responding to commands	Check that the relay is shown as online on the dashboard. If offline, check power and proximity.
Factory reset not working on indoor sensor	Press the Reset button 3 times in quick succession, no more than 30 seconds apart. The counter clears once the sensor finishes a reading cycle. The sensor has no LED — check the Pending Devices tab to confirm.
Factory reset not working on outdoor sensor	Hold Reset down while switching the sensor On , and keep holding for the full 3 seconds — releasing early cancels the reset. Watch for 6 fast LED flashes to confirm.
Can't access gardenmonitor.local	Use the IP address shown on the Garden Monitor's display instead. Check that your phone/computer is on the same WiFi network as the Garden Monitor.